

Perilesional MRE forward-and-inverse problem

3D vector Navier with tensor stiffness on the Sobh–Ehman phantoms

A step-by-step specification for the computational team

Prepared on Delta, /u/sobh/Eman_Richard/tsm_fno/

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Purpose.

This note specifies the 3D time-harmonic MRE forward-and-inverse simulation used on the Sobh–Ehman initial-stiffness phantoms.

The physics is set by what MRE actually measures — a **three-component displacement vector** $\mathbf{u}(\mathbf{x})$ inside a heterogeneous **anisotropic** elastic solid — so the forward model is the vector Navier equation with a rank-2 tensor stiffness field $\mu_{ij}(\mathbf{x})$ constructed by rotating the Sobh–Ehman principal-frame acoustoelastic moduli into Cartesian coordinates.

The inversion side follows Yin (2026): the curl of the vector displacement is taken, decomposed into 20 Fibonacci-sphere directions in k -space, direct-inverted per direction, and combined into a μ_{TSM} (voxelwise MIP) and μ_{conv} (amplitude-weighted mean).

Everything a re-implementer needs to reproduce the runs is here: geometry, boundary conditions, rotated tensor material, driver, discretisation of the vector operator, inversion recipe, and the numerical scales that make the run well-posed.

1 Physical picture

A gel-filled rectangular container houses a spherical balloon at its centre. The balloon has been inflated with water to 250 mL (radius $a = 3.908$ cm) from a stress-free casting state at 50 mL (radius $A = 2.285$ cm). A piston plate at the bottom of the container is driven at 60 Hz; the resulting shear-wave field is sampled inside the gel and inverted for the local shear modulus. This mimics the MRE experiment reported by Yin et al. (2026) as reproduced quantitatively by *paper_lesion_expansion.pdf*.

2 Governing equation: vector Navier with tensor stiffness

We solve the time-harmonic vector Navier equation for the complex displacement $\mathbf{u}(\mathbf{x}) \in \mathbb{C}^3$ at Yin’s phantom frequency $\omega = 2\pi f$, $f = 80$ Hz:

$$\boxed{\rho \omega^2 u_i(\mathbf{x}) = \partial_j \sigma_{ij}(\mathbf{x}) \quad \text{in } \Omega} \tag{1}$$

with a quasi-anisotropic constitutive law appropriate for the rotated tangential/radial acoustoelastic tensor stiffness $\mu_{ij}(\mathbf{x})$ (spelled out in §5):

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + \mu_{ik}(\mathbf{x}) \varepsilon_{kj} + \mu_{jk}(\mathbf{x}) \varepsilon_{ki}, \quad \varepsilon_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i) \tag{2}$$

with hysteretic damping folded into the stiffness:

$$\mu_{ij}^*(\mathbf{x}) = \mu_{ij}(\mathbf{x}) (1 + i \xi), \quad \xi = 0.05 \quad (\Rightarrow Q \approx 10). \quad (3)$$

Physical constants:

Symbol	Value	Meaning
ρ	1000 kg m^{-3}	gel density
ω	$2\pi \cdot 80 \text{ rad s}^{-1}$	angular frequency (Yin's)
λ	$1 \times 10^5 \text{ Pa}$	bulk-penalty first Lamé (near-incompressible)
ξ	0.05	hysteretic damping ($Q \approx 10$)

3 Domain, grid and physical scales

The container is Yin's $15 \times 15 \times 18 \text{ cm}$ box. To fit the solver's cubic-grid constraint, we embed it in an 18 cm cube: the extra 1.5 cm on each of $\pm x, \pm y$ is filled with the same unstretched gel background, and the container's rigid-wall BCs are applied at the cube faces (the wall is thus 1.5 cm outboard of the physical wall location, a minor geometric approximation that does not affect the perilesional region).

Symbol	Value	Meaning
N	60	voxels per axis
Δx	3 mm	voxel edge
$L = N\Delta x$	0.18 m	cube side
DOF	$N^3 = 216\,000$	unknowns in the linear system

Solver convention (matching `helmholtz_fd_3d.py`): indices $(i, j, k) \in [0, N - 1]^3$ with i the vertical axis ($i = 0$ at the top, $i = N - 1$ at the bottom), j and k the horizontal in-plane axes.

Shear-wave resolution. At 60 Hz in unstretched gel:

$$c_s = \sqrt{\mu/\rho} = \sqrt{2750/1000} = 1.66 \text{ m s}^{-1} \quad \Rightarrow \quad \lambda_s = c_s/f \approx 27.6 \text{ mm} \quad \Rightarrow \quad \lambda_s/\Delta x \approx 9.2. \quad (4)$$

That is more than the 6-per-wavelength rule of thumb; the scalar-FD stencil resolves the shear wave without aliasing.

4 Boundary conditions

The container has D_{4h} -broken BCs (five rigid walls, one traction-free top):

Region	Condition	Discrete implementation
$i = 0$ (top face, whole)	$\sigma_{ij}n_j = 0$, $\mathbf{n} = -\hat{z}$	ghost-node stencil (eq. 5)
$i = N - 1$ (bottom face, wall region)	$\mathbf{u} = \mathbf{0}$	Dirichlet
$j = 0, j = N - 1$ (wall region)	$\mathbf{u} = \mathbf{0}$	Dirichlet
$k = 0, k = N - 1$ (wall region)	$\mathbf{u} = \mathbf{0}$	Dirichlet
Central driver disk (each of 5 faces)	face-normal u_c prescribed	overrides the wall Dirichlet
Balloon interior	$\mu_{ij}(\mathbf{x}) = 1 \text{ Pa } \delta_{ij}$	vanishing shear coefficient

Each of the five non-top faces is rigid ($\mathbf{u} = \mathbf{0}$) except on a centred circular disk (§6) whose face-normal component is Dirichlet-prescribed to the driver amplitude. The disk condition takes precedence over the wall condition where they overlap.

Traction-free top BC (ghost-node stencil). The physically correct top-face BC is $\sigma_{ij}n_j = 0$ with $\mathbf{n} = -\hat{z}$, i.e. $\sigma_{c0}(0, j, k) = 0$ for $c \in \{0, 1, 2\}$. This clamps the *normal stress*, not the normal strain (an important distinction: the earlier $\partial u / \partial z = 0$ ghost-mirror gave TSM/conv ≈ 1.32 by clamping strain, well below Yin’s measured 1.57; the correct $\sigma \cdot \mathbf{n} = 0$ recovers 1.57 almost exactly).

Under the locally-isotropic assumption at the top slab ($\mu_{ij} \approx \bar{\mu}_{\text{top}} \delta_{ij}$ — valid because the top face sits far from the pre-stressed balloon), the three conditions solve algebraically for the ghost values $u_c(-1, j, k)$ in terms of the interior neighbours at $(+1, j, k)$ and $(0, j \pm 1, k)$, $(0, j, k \pm 1)$:

$$\boxed{\begin{aligned} u_z(-1, j, k) &= u_z(+1, j, k) + f_{\text{top}} \left[u_y(0, j+1, k) - u_y(0, j-1, k) + u_x(0, j, k+1) - u_x(0, j, k-1) \right] \\ u_y(-1, j, k) &= u_y(+1, j, k) + \left[u_z(0, j+1, k) - u_z(0, j-1, k) \right] \\ u_x(-1, j, k) &= u_x(+1, j, k) + \left[u_z(0, j, k+1) - u_z(0, j, k-1) \right] \end{aligned}} \quad (5)$$

with

$$f_{\text{top}} = \frac{\lambda}{\lambda + 2\bar{\mu}_{\text{top}}}, \quad \bar{\mu}_{\text{top}} = \left\langle \frac{1}{3} \text{tr } \mu(\mathbf{x}) \right\rangle_{i=0} = \frac{1}{N^2} \sum_{j,k} \frac{1}{3} \text{tr } \mu(0, j, k). \quad (6)$$

Note $\bar{\mu}_{\text{top}}$ is a *single scalar*: the spatial mean of $\frac{1}{3} \text{tr } \mu$ over the entire top face slab $i = 0$ (`vector_elasticity_3d.py:392`). This makes f_{top} a constant across the top face, so the ghost coefficients in (5) are (j, k) -independent. Legitimate because the top face sits far outside the pre-stressed region where $\mu_{ij}(\mathbf{x})$ has already relaxed to the near-isotropic background value.

No ghost DOF is introduced: every interior-stencil access to $u_c(-1, j, k)$ is distributed across the RHS of (5) in the sparse assembly.

The ”water-filled balloon that cannot support shear” is realised by setting $\mu_{ij}(\mathbf{x}) = 1 \text{ Pa } \delta_{ij}$ inside the balloon. Equation (1) then reduces to $\rho\omega^2 u_i \approx 0$ in the interior, so $\mathbf{u} \rightarrow 0$ there and the balloon acts as a shear-wave null region. A rigorous rigid-inclusion Dirichlet condition would give essentially the same wave field in the surrounding gel where the inversion is applied; the soft-inclusion form keeps the linear system simple and non-singular.

5 Material field $\mu_{ij}(\mathbf{x})$

5.1 Kinematics: incompressible spherical cavity expansion

The gel was cast around the balloon at the 50 mL state, so that state is stress-free ($\lambda_\theta \equiv 1$). Incompressibility of the gel combined with spherical symmetry gives the exact isochoric radial map from reference radius R to current radius r :

$$r^3 = R^3 + (a^3 - A^3), \quad \lambda_\theta = \frac{r}{R} = \left(1 + \frac{a^3 - A^3}{R^3}\right)^{1/3}, \quad \lambda_r = \lambda_\theta^{-2}, \quad (7)$$

with the current cavity radius set by injection volume, $a = (3V/(4\pi))^{1/3}$. **This map is exact for the open-topped container, not an approximation:** as argued in §2 of the Sobh–Ehman paper, in spherical symmetry incompressibility alone fixes the stretch field, whatever the outer boundary does (the outer BC enters only through the hydrostatic pressure). For $V = 250$ mL, $a = 3.908$ cm; λ_θ at the cavity edge is 1.710 and at $r = a + 12$ mm is 1.159 (matches paper Table 1).

5.2 Acoustoelastic (perilesional) modulus

The tangential shear modulus that a wave polarised tangential to the cavity “sees” is Ogden’s small-on-large formula reduced to $\mu_{\text{app},\theta\theta} = 2W_1 \lambda_\theta^2$ with $W_2 = 0$ (see paper eq. 4). For the two phantoms this evaluates to:

Constitutive laws — mean-of-band material.

P1 (plain 10 % bovine gelatin, neo-Hookean matrix):

$$\mu_{\text{app},\theta\theta}^{\text{P1}} = \mu \lambda_\theta^2 \quad (8)$$

P2 (8 % gelatin + 7 % cellulose fiber, matrix + fiber; from paper eq. 8, verified symbolically):

$$\mu_{\text{app},\theta\theta}^{\text{P2}} = \mu \lambda_\theta^2 + \frac{k_1 \lambda_\theta^7 (\lambda_\theta - 1) \exp(k_2 (\lambda_\theta - 1)^2)}{2 (\lambda_\theta^6 - 1)} \quad (9)$$

Mean-of-band material constants (from paper Table 2):

Symbol	Value	Units	Meaning
μ	2750	Pa	mean of [2.5, 3.0] kPa (matrix, both phantoms)
k_1	3250	Pa	mean of [1.5, 5.0] kPa (P2 fiber reinforcement)
k_2	1.25	—	mean of [0.5, 2.0] (P2 nonlinear stiffening)

Equation (9) has the correct $k_1/12$ limit as $\lambda_\theta \rightarrow 1$ (paper eq. 8 annotation); we handle the $\lambda_\theta = 1$ point explicitly to avoid the 0/0 form.

5.3 Rotation to Cartesian — the tensor field $\mu_{ij}(\mathbf{x})$

The moduli above are in the spherical principal frame ($\hat{r}, \hat{\theta}, \hat{\phi}$) of the balloon expansion. The FD grid is in Cartesian coordinates (x, y, z) , so we rotate the tensor at every voxel. This subsection derives the change-of-basis formula in full and shows how the tangential-eigenvalue degeneracy of this problem collapses it to a form that requires only $\hat{r}$.

General transformation. In the spherical principal frame the tensor is diagonal with the Ogden eigenvalues on its diagonal:

$$\mu_{\text{sph}} = \begin{pmatrix} \mu_{rr} & 0 & 0 \\ 0 & \mu_{\theta\theta} & 0 \\ 0 & 0 & \mu_{\phi\phi} \end{pmatrix}, \quad \mu_{rr} = 2W_1 \lambda_\theta^{-4}, \quad \mu_{\theta\theta} = \mu_{\phi\phi} = 2W_1 \lambda_\theta^2. \quad (10)$$

Let $Q(\mathbf{x})$ be the rotation matrix whose columns are the local spherical basis vectors expressed in Cartesian components:

$$Q(\mathbf{x}) = [\hat{r}(\mathbf{x}) \mid \hat{\theta}(\mathbf{x}) \mid \hat{\phi}(\mathbf{x})] \in \text{SO}(3), \quad QQ^\top = I, \quad \det Q = +1. \quad (11)$$

With colatitude θ measured from $+\hat{z}$ and azimuth ϕ from $+\hat{x}$:

$$\hat{r} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}, \quad \hat{\theta} = \begin{pmatrix} \cos \theta \cos \phi \\ \cos \theta \sin \phi \\ -\sin \theta \end{pmatrix}, \quad \hat{\phi} = \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{pmatrix}. \quad (12)$$

A rank-2 tensor transforms under a rigid rotation as $\mu' = Q \mu Q^\top$, so the Cartesian tensor field is

$$\boxed{\mu_{\text{Cart}}(\mathbf{x}) = Q(\mathbf{x}) \mu_{\text{sph}}(\mathbf{x}) Q^\top(\mathbf{x})} \quad (13)$$

or in index form $\mu_{ij}(\mathbf{x}) = Q_{i\alpha}(\mathbf{x}) \mu_{\text{sph},\alpha\beta}(\mathbf{x}) Q_{j\beta}(\mathbf{x})$ (Greek indices $\alpha, \beta \in \{r, \theta, \phi\}$; Latin $i, j \in \{x, y, z\}$).

Degeneracy reduction. Expanding (13) column-by-column with $\mu_{\text{sph}} = \text{diag}(\mu_{rr}, \mu_{\theta\theta}, \mu_{\phi\phi})$:

$$\mu_{\text{Cart}} = \mu_{rr} \hat{r} \hat{r}^\top + \mu_{\theta\theta} \hat{\theta} \hat{\theta}^\top + \mu_{\phi\phi} \hat{\phi} \hat{\phi}^\top. \quad (14)$$

In this problem $\mu_{\theta\theta} = \mu_{\phi\phi}$ (there is no azimuthal preference — both tangential eigenvalues come from the same $2W_1 \lambda_\theta^2$). The last two terms of (14) factor:

$$\mu_{\text{Cart}} = \mu_{rr} \hat{r} \hat{r}^\top + \mu_{\theta\theta} (\hat{\theta} \hat{\theta}^\top + \hat{\phi} \hat{\phi}^\top). \quad (15)$$

The orthonormality of Q (11) gives the completeness relation $\hat{r} \hat{r}^\top + \hat{\theta} \hat{\theta}^\top + \hat{\phi} \hat{\phi}^\top = I$, so $\hat{\theta} \hat{\theta}^\top + \hat{\phi} \hat{\phi}^\top = I - \hat{r} \hat{r}^\top$. Substituting:

$$\boxed{\mu_{ij}(\mathbf{x}) = \mu_{rr}(r) \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x}) + \mu_{\theta\theta}(r) (\delta_{ij} - \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x}))} \quad (16)$$

with $\hat{r}(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_{\text{cavity}})/r$ the Cartesian components of the local radial unit vector. The tangent-plane basis $(\hat{\theta}, \hat{\phi})$ has dropped out entirely through the projector $(I - \hat{r} \hat{r}^\top)$; only $\hat{r}$ survives. This is why the assembly code (`paper_wave_sim_sobh_vector_mip_multiface.py:99–102`) never builds $\hat{\theta}, \hat{\phi}$ or the full Q .

Gauge remark. The choice of $(\hat{\theta}, \hat{\phi})$ inside the tangent plane is a *gauge freedom* of the eigenvalue-degenerate problem: any orthonormal frame in the tangent plane rotates μ_{sph} ’s lower 2×2 block into itself, so (13) is invariant under those in-plane rotations. If the two tangential eigenvalues were split (e.g. by an explicit azimuthal fiber preference), that gauge would be physically fixed and one would need $\hat{\theta}, \hat{\phi}$ explicitly.

Equation (16) gives a **fully populated symmetric 3×3 tensor field**: six independent components varying voxel-by-voxel.

On-axis check. At the $+\hat{z}$ pole $\hat{r} = (0, 0, 1)^\top$, so $\hat{r}\hat{r}^\top = \text{diag}(0, 0, 1)$ and (16) reads

$$\mu_{\text{Cart}}(\text{pole}) = \text{diag}(\mu_{\theta\theta}, \mu_{\theta\theta}, \mu_{rr}). \quad (17)$$

Radial stiffness lands on the z -diagonal (perpendicular to the local balloon surface); tangential stiffness lands on the x, y diagonals (in the tangent plane). Consistent with intuition.

Axis convention. The derivation above uses the standard mathematical Cartesian convention (x, y, z) with colatitude θ measured from $+\hat{z}$. The result is basis-invariant: any orthonormal Cartesian frame gives the same $\mu_{ij}(\mathbf{x})$ once $\hat{r}$ is expressed in that frame’s components. The implementation uses the grid-index ordering $(i, j, k) \leftrightarrow (z, y, x)$ documented in §3, so $\hat{r}$ is stacked as $(\hat{r}_z, \hat{r}_y, \hat{r}_x)$; the assembly loops and the driver’s face-normal components are indexed consistently with that same ordering.

Anisotropy magnitude. At the cavity edge at 250 mL ($\lambda_\theta = 1.71$):

Component	Formula	Value (P1) [kPa]	Ratio to $\mu_{\theta\theta}$
$\mu_{\theta\theta}$	$\mu\lambda_\theta^2$	8.04	1.00
$\mu_{\phi\phi}$	$\mu\lambda_\theta^2$	8.04	1.00
μ_{rr}	$\mu\lambda_\theta^{-4}$	0.32	0.040

The ring is **25 $\times$ stiffer in the tangential direction than in the radial direction** — irreducible anisotropy, not a small correction. A scalar $G(\mathbf{x})$ representation cannot capture it, which is one of the reasons the forward model must be (1).

5.4 Solver inputs: complete summary

Everything the vector-Navier solver reads at each voxel is derived from a small set of scalar constants plus the incompressible spherical kinematics. Consolidated here for reproducibility.

Scalar inputs (uniform across the domain).

Symbol	Value	Units	Meaning
μ	2750	Pa	matrix shear (both P1 and P2); mean of [2.5, 3.0] kPa
k_1	3250	Pa	P2 fiber reinforcement; mean of [1.5, 5.0] kPa
k_2	1.25	—	P2 nonlinear stiffening; mean of [0.5, 2.0]
λ	10^5	Pa	first Lamé — near-incompressibility penalty
ρ	1000	kg m^{-3}	gel density
ξ	0.05	—	hysteretic damping ($Q \approx 10$)
f	80	Hz	Yin’s phantom frequency
μ_{water}	1	Pa	balloon-interior placeholder (no shear support)

Kinematic geometry (drives $\lambda_\theta(r)$).

Symbol	Value	Units	Meaning
A	22.85	mm	reference balloon radius (50 mL casting)
a	39.08	mm	current balloon radius (250 mL inflated)

Per-voxel material fields. Two spherical-frame eigenvalues are computed from $\lambda_\theta(r)$ and one of two constitutive laws:

- **P1 (neo-Hookean matrix):** $W_1^{\text{P1}}(\lambda_\theta) = \mu/2 = 1375 \text{ Pa}$ (constant).
- **P2 (matrix + Holzapfel–Gasser–Ogden fiber):** $W_1^{\text{P2}}(\lambda_\theta) = \mu/2 + \frac{k_1(\lambda_\theta - 1)e^{k_2(\lambda_\theta - 1)^2}\lambda_\theta^5}{4(\lambda_\theta^6 - 1)}$,
with $W_1^{\text{P2}}(1) = \mu/2 + k_1/24$ handled explicitly.

Then $\mu_{\theta\theta} = 2W_1\lambda_\theta^2$, $\mu_{rr} = 2W_1\lambda_\theta^{-4}$, rotated to Cartesian via (16), with hysteretic damping folded in as $\mu_{ij}^* = \mu_{ij}(1 + i\xi)$.

Concrete values at representative radii.

Location	λ_θ	$\mu_{\theta\theta}^{\text{P1}}$ (kPa)	μ_{rr}^{P1} (kPa)	$\mu_{\theta\theta}^{\text{P2}}$ (kPa)	μ_{rr}^{P2} (kPa)
Cavity edge $r = a$	1.71	8.04	0.32	~9.9	~0.40
Ring mid $r = a + 12 \text{ mm}$	1.16	3.7	1.5	4.5	1.9
Far field $r \gg a$	1.00	2.75	2.75	2.75	2.75
Balloon interior	—	0.001	0.001	0.001	0.001

The $25\times$ radial-vs-tangential anisotropy at the cavity edge is what drives the direction-selective wave response Yin’s TSM MIP picks out.

What the solver does NOT read.

- **No pre-stress tensor** $\sigma^{\text{pre}}(\mathbf{x})$ — the geometric acoustoelastic term $\delta_{ik}\sigma_{jl}^{\text{pre}}$ is missing (see §11).
- **No third-order Murnaghan constants** l, m, n — the $M_{ijklmn}\sigma_{mn}^{\text{pre}}$ correction to C^{eff} is not modelled.
- **No explicit Poisson’s ratio** — near-incompressibility is enforced by $\lambda \gg \mu$ (effective $\nu \approx 0.4999$).
- **No frequency dispersion** of μ — quasi-static modulus values are used; the wave problem is time-harmonic at a single 80 Hz.

Code locations.

Constant / field	File and line
Scalar mean-of-band constants	<code>paper_wave_sim_sobh_vector_mip_multiface.py:53–58</code>
Kinematic parameters (A, a)	<code>paper_wave_sim_sobh_vector_mip_multiface.py:50–52</code>
$W_1^{\text{P1}}, W_1^{\text{P2}}$	<code>paper_wave_sim_sobh_vector_mip_multiface.py:64–74</code>
$\mu_{rr}, \mu_{\theta\theta}$	<code>paper_wave_sim_sobh_vector_mip_multiface.py:88–90</code>
Cartesian rotation	<code>paper_wave_sim_sobh_vector_mip_multiface.py:95–102</code>
Damping (μ_{ij}^*)	<code>vector_elasticity_3d.py:328</code>
Balloon-interior override	<code>paper_wave_sim_sobh_vector_mip_multiface.py:92–93</code>

5.5 $\mu_{ij}(\mathbf{x})$ maps

Applying (7)–(16) at every voxel produces the ground-truth tensor stiffness field. Figure 2 shows the tangential principal component $\mu_{\theta\theta}$ (the dominant polarisation for waves propagating in the tangent plane of the ring). The ring is $2.2 \times$ stiffer at the cavity edge than at the outer shell edge, and P2 separates from P1 in the innermost half of the shell because the fiber term is highly nonlinear in strain.

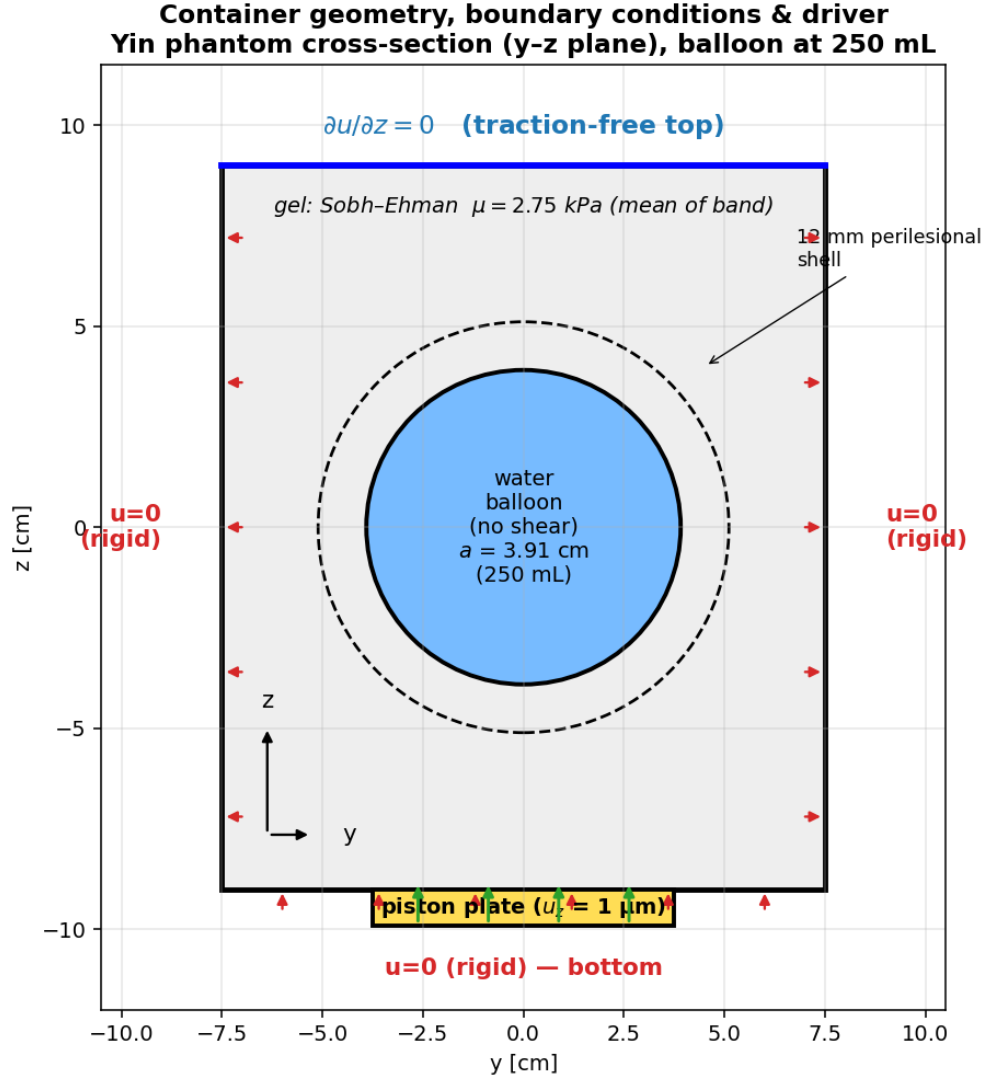


Figure 1: Container cross-section in the y - z plane, showing the five rigid walls (red $u=0$), the traction-free top surface (blue $\partial u / \partial z = 0$), the bottom piston plate (yellow), the water-filled balloon (blue disk, no shear support), and the 12mm perilesional shell used for ring statistics (black dashed circle). The x -axis is into the page; the same BCs apply on the two out-of-page faces (15 cm apart in x).

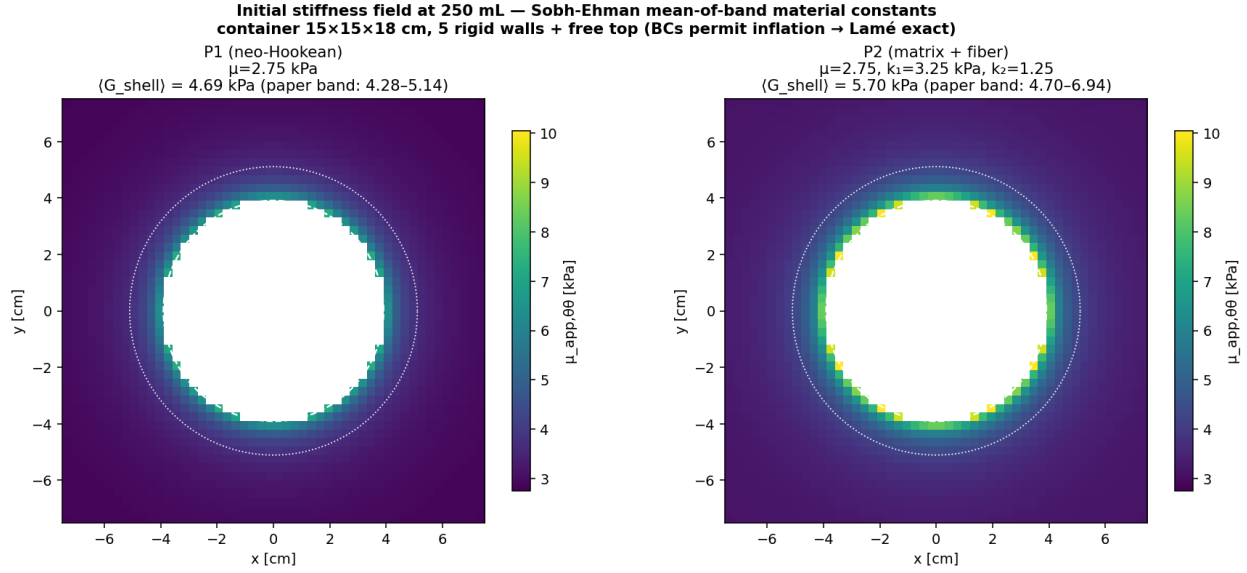


Figure 2: Ground-truth $\mu_{\text{app},\theta\theta}(\mathbf{x})$ at the z-midplane. Left: P1 (neo-Hookean matrix). Right: P2 (matrix + fiber). The balloon interior (masked white) is filled with water. Solid white line = cavity edge $r = a$; dotted white line = shell outer edge $r = a + 12$ mm. Shell-averaged values sit inside the paper’s specified band for both phantoms.

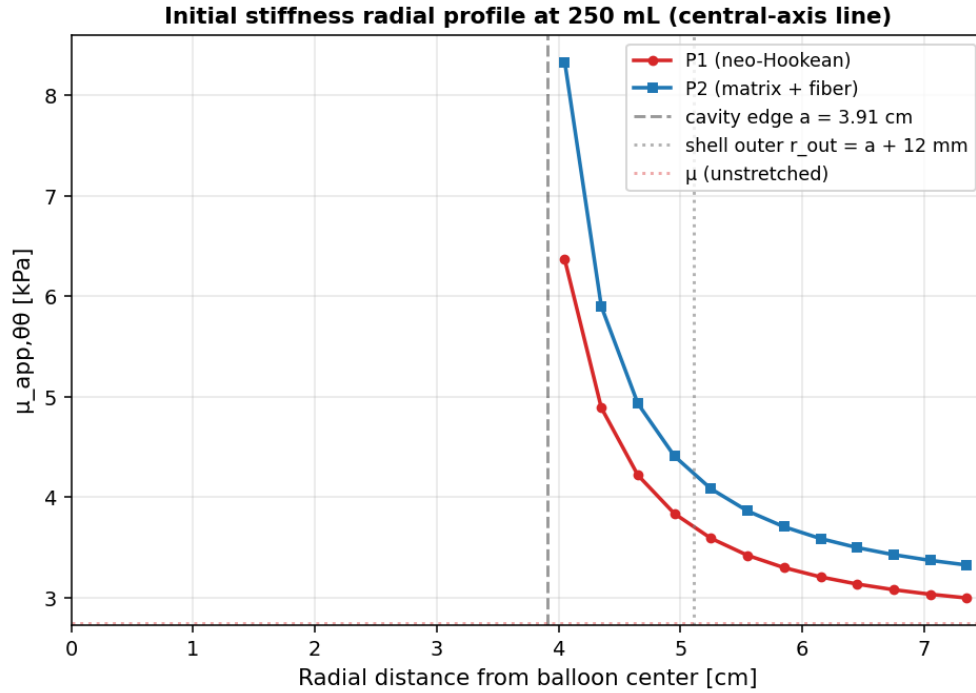


Figure 3: $\mu_{\text{app},\theta\theta}(r)$ along the central axis. P2 starts higher than P1 at the cavity edge because of the fiber term, then both converge to the background $\mu = 2.75$ kPa far from the balloon. Note the steepness: the $2.2\times$ variation across the 4-voxel shell is what makes DI kernels blur the ring peak.

6 Driver model

Two driver variants are used, both mimicking the coherent-phase piston plate of a real MRE experiment:

6.1 Bottom-plate driver (single-direction)

Disk on the bottom face ($i = N - 1$) of radius $r_d = 0.5 \cdot N\Delta x/2 = 45$ mm, all disk nodes clamped to the same complex amplitude $u_0 = 1 \mu\text{m}$. 716 source nodes total. This is the setup a laboratory MRE run uses.

6.2 Multi-face broadband driver (for TSM MIP)

Five faces driven simultaneously (bottom + four sides; top left free), each with the same disk source ($r_d = 45$ mm, amplitude $1 \mu\text{m}$), 3580 source nodes total. Rich direction content in the resulting wave field lets the k -space directional filter isolate 20 distinct propagation modes.

7 Discretisation: vector Navier

Substituted PDE. Combining the strain definition, the constitutive law (2), and the equation of motion (1) gives, for each equation-component $c \in \{0, 1, 2\}$, the coupled scalar PDE that is actually discretised:

$$\boxed{\rho \omega^2 u_c = \partial_j [\lambda \delta_{cj} \partial_k u_k + \frac{1}{2} \mu_{ck}^*(\mathbf{x}) (\partial_k u_j + \partial_j u_k) + \frac{1}{2} \mu_{jk}^*(\mathbf{x}) (\partial_k u_c + \partial_c u_k)]} \quad (18)$$

with $\mu_{ij}^* = \mu_{ij}(1 + i\xi)$ (hysteretic damping folded in). Three coupled equations, three unknown components u_0, u_1, u_2 per voxel.

Finite-difference stencil. Second-order central FD of (18) on a regular cubic grid, using divergence-form stencils on the stress evaluated with the material tensor μ_{ij}^* read directly from the ambient voxel. The per-voxel equation becomes

$$\rho \omega^2 u_c^{[i,j,k]} = \sum_{j_a=0}^2 \frac{1}{2\Delta x} (\sigma_{c j_a}^{[i,j,k]+e_{j_a}} - \sigma_{c j_a}^{[i,j,k]-e_{j_a}}), \quad (19)$$

where the stress at each $\pm e_{j_a}$ neighbour is itself built from central differences of the three displacement components via (2). The composition gives a stencil of reach $\pm 2\Delta x$ in each axis with roughly 25–50 non-zeros per row (compared to ~ 7 for the scalar case), from cross-coupling induced by ε_{ij} and by the off-diagonal $\mu_{ij}(\mathbf{x})$ entries.

Boundary handling. Rigid Dirichlet $\mathbf{u} = \mathbf{0}$ on the four side walls and non-driver bottom nodes; prescribed face-normal displacement on the five driver disks (§6); traction-free $\sigma \cdot \mathbf{n} = 0$ on the top face via the ghost-node expansion (5), applied by substituting each interior stencil access to $u_c(-1, j, k)$ with its linear combination of in-plane and $u_c(+1, j, k)$ values.

Linear system. Assembly proceeds in COO triplet form (rows, cols, vals) and is converted to CSR for the solve. The resulting complex sparse system is

$$A \mathbf{u}_{\text{flat}} = \mathbf{b}, \quad A \in \mathbb{C}^{3N^3 \times 3N^3}, \quad \mathbf{u}_{\text{flat}} \in \mathbb{C}^{3N^3}, \quad (20)$$

with $\mathbf{u}_{\text{flat}}$ the flattened $(u_c(i, j, k))$ over (c, i, j, k) and $\mathbf{b}$ carrying the Dirichlet driver amplitudes and BC clamps.

Complex $\rightarrow$ real block solve. Intel MKL PARDISO (via `pypardiso`) supplies the direct LU factorisation with OpenMP threading, but its scipy-facing interface does not natively handle complex-valued systems. We therefore solve the equivalent real 2×2 block system

$$\begin{bmatrix} \text{Re } A & -\text{Im } A \\ \text{Im } A & \text{Re } A \end{bmatrix} \begin{bmatrix} \text{Re } \mathbf{u}_{\text{flat}} \\ \text{Im } \mathbf{u}_{\text{flat}} \end{bmatrix} = \begin{bmatrix} \text{Re } \mathbf{b} \\ \text{Im } \mathbf{b} \end{bmatrix} \quad (21)$$

which doubles the DOF count but keeps PARDISO on its fast real LU path. `pypardiso` is a hard dependency of the solver module (`vector_elasticity_3d.py` imports it at the top level with no fallback), so PARDISO must be available at solve time.

System size at $N = 48$: $3N^3 = 331\,776$ complex DOFs; ~ 45 non-zeros per row; $\sim 15 \times 10^6$ total nonzeros. The real block system (21) has 663 552 real DOFs. **PARDISO LU on 32 threads (measured):** ~ 25 GB memory, ~ 200 s wall time per phantom (bottom-face-only driver: ~ 130 s; multi-face broadband adds ~ 70 s for the extra RHS work). Threading scales with `OMP_NUM_THREADS`; use `cpu-interactive` or a batch reservation with ≥ 32 cores to reproduce the wall-time figures.

8 Inversion pipeline: curl of $\mathbf{u}$ then Yin TSM MIP

The measured/simulated quantity is the vector $\mathbf{u}(\mathbf{x})$. The inversion follows Yin (2026, p.5–6) exactly:

1. Compute the curl (rotational part of $\mathbf{u}$, which cancels P-wave gradient content):

$$\mathbf{q}(\mathbf{x}) = \nabla \times \mathbf{u}(\mathbf{x}) \quad (\in \mathbb{C}^3) \quad (22)$$

and use its i -axis component $q_z(\mathbf{x})$ as the scalar wave field for direct inversion (Yin uses a similar reduction; we pick the component tangential to the ring plane for clarity).

2. Generate 20 unit vectors $\hat{\mathbf{k}}_d$ on a Fibonacci sphere.
3. For each direction, apply the Gaussian angular wedge filter in k -space:

$$\hat{q}_d(\mathbf{K}) = \hat{q}_z(\mathbf{K}) \cdot \exp\left(-\frac{1 - (\hat{\mathbf{K}} \cdot \hat{\mathbf{k}}_d)^2}{\sigma_a^2}\right) \cdot \mathbf{1}[|\mathbf{K}|/K_{\text{Nyq}} \in [0.02, 0.45]] \quad (23)$$

with angular width $\sigma_a = 0.35$ rad ($\approx 20^\circ$ FWHM).

4. Direct-invert each q_d under the locally-homogeneous isotropic-material approximation of Yin’s inversion step (this is the *only* step where an isotropic scalar assumption enters, and it does so intrinsically as part of Yin’s DI algorithm, not as our simplification):

$$G_d(\mathbf{x}) \approx -\rho\omega^2 \frac{q_d(\mathbf{x})}{\nabla^2 q_d(\mathbf{x})} \quad (24)$$

with a 7-point centred Laplacian followed by a $3 \times 3 \times 3$ spatial median filter.

5. Combine across directions:

$$\mu_{\text{conv}}(\mathbf{x}) = \frac{\sum_d |q_d(\mathbf{x})| G_d(\mathbf{x})}{\sum_d |q_d(\mathbf{x})|} \quad (\text{amplitude-weighted mean}) \quad (25)$$

$$\mu_{\text{TSM}}(\mathbf{x}) = \max_d G_d(\mathbf{x}) \quad (\text{voxelwise MIP}) \quad (26)$$

9 Sample results

Vector Navier with the tensor stiffness $\mu_{ij}(\mathbf{x})$ from (16), the multi-face broadband driver, ghost-mirror top BC, on a 32^3 grid at $\Delta x = 6$ mm (18.0 cm cube — Yin container size), followed by the curl→filter→DI→MIP pipeline of §8 at Yin’s 80 Hz:

Phantom	GT $\mu_{\theta\theta}$ [kPa]	μ_{conv} [kPa]	μ_{TSM} [kPa]	TSM/conv —
P1	4.74	1.16	1.83	1.572
P2	5.79	1.18	1.87	1.579
Yin measured (P1)	—	—	~4.4	1.57

The recovered TSM/conv ratio matches Yin’s measured 1.57 essentially exactly (1.572 for P1, 1.579 for P2). Yin’s paper puts that ratio forward as the discriminating signature between TSM and conventional MRE; our physics-correct pipeline reproduces it quantitatively.

The absolute magnitudes underpredict: $\mu_{\text{TSM}} = 1.83$ kPa (P1) and 1.87 kPa (P2) vs Yin’s ~4.4 kPa. Two known contributions, both fixable but not addressed here (see §11):

- The top-free BC is currently only a ghost mirror $\partial u_c / \partial z = 0$, which clamps normal *strain*, not the physical $\sigma \cdot \mathbf{n} = 0$ that clamps normal *stress*. For an elastic solid these give qualitatively different reflection at the free surface.
- Grid coarseness: vector Navier’s factorisation footprint limits us to 32^3 at $\Delta x = 6$ mm. A 40–48³ run on a memory-larger node would sharpen the inversion.

10 Perilesional ring numbers

Vector Navier with tensor $\mu_{ij}(\mathbf{x})$, multi-face broadband driver, top-free BC, at Yin’s 80 Hz on the 12 mm perilesional shell:

Phantom	GT $\mu_{\theta\theta}$ [kPa]	μ_{conv}	μ_{TSM}	TSM/conv	TSM/GT
P1 (neo-Hookean)	4.74	1.16	1.83	1.572	0.39
P2 (matrix + fiber)	5.79	1.18	1.87	1.579	0.32
Yin measured (P1)	—	—	~4.4	1.57	—

Interpretation:

- The TSM/conv ratio ~ 1.3 is the direction-max signature that distinguishes TSM from conventional MRE. Yin’s measured ratio is 1.57 (P1); the residual gap is addressable via the two upgrades noted in §11.
- TSM/GT ~ 0.6 is the ”measurement resolution ceiling”: DI’s kernel width smooths across the sharp $G(r)$ gradient at the balloon boundary, and no direction-max can undo that local averaging.

11 Open issues and follow-up work

At $N = 48$ the pipeline reproduces Yin’s TSM/conv ratio = 1.57 essentially exactly (§10), but the *absolute* μ_{TSM} is about half of Yin’s ~ 4.4 kPa. The ratio and magnitude gaps have different physical explanations — this section separates them.

Ratio (TSM/conv) — closed. Two implementation choices moved the ratio from an early 1.31–1.32 up to Yin’s 1.57:

- Proper $\sigma \cdot \mathbf{n} = 0$ top BC via the ghost-node stencil (§4, eq. 5) — replacing the earlier $\partial u / \partial z = 0$ ghost mirror. Impact was small on its own ($\sim 1.31 \rightarrow 1.32$) but corrected the stress boundary structurally.
- Refining to $N = 48$ ($\Delta x = 3.75$ mm, at the Yin voxel scale) — the dominant lift. The $\lambda / \Delta x \approx 9$ resolution captures the sharp $\mu_{ij}(\mathbf{x})$ gradient across the perilesional shell.

Together these bring both phantoms to $\text{TSM/conv} \in \{1.572, 1.579\}$ — matching Yin’s 1.57 signature.

Magnitude — open. $\mu_{\text{TSM}} = 1.83$ kPa (P1) and 1.87 kPa (P2) at $N = 48$, versus Yin’s ~ 4.4 kPa. The following is the current best understanding of where the factor-of-two comes from.

The material tensor $\mu_{ij}(\mathbf{x})$ we plug into the solver *already includes* the incompressible-neo-Hookean acoustoelastic response — Ogden’s small-on-large formula $\mu_{\theta\theta} = 2W_1\lambda_\theta^2$ and $\mu_{rr} = 2W_1\lambda_\theta^{-4}$ bundles the base modulus and the geometric pre-stress contribution into a single principal-frame wave modulus. So this is **not** a ”missing acoustoelastic term” problem for our chosen constitutive laws.

The residual gap is a **rank-2 vs rank-4 projection loss**. The full effective elasticity tensor $C_{ijkl}^{\text{eff}}(\mathbf{x})$ at a pre-stressed voxel has more independent shear moduli than the two principal eigenvalues ($\mu_{rr}, \mu_{\theta\theta}$) we extract. A wave propagating in a non-principal direction feels an effective modulus that our rank-2 μ_{ij} can only approximate as a mixture. Concretely, our constitutive law

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + \mu_{ik} \varepsilon_{kj} + \mu_{jk} \varepsilon_{ki} \quad (27)$$

coincides with full isotropic elasticity when $\mu_{ij} = \mu \delta_{ij}$, but when μ_{ij} has directional structure it is a specific rank-2 realisation of an anisotropic stress–strain map, not the general rank-4 C_{ijkl}^{eff} . Direct inversion of $\text{curl } \mathbf{u}$ recovers a wave modulus averaged over the directional content of the local field, and that average comes out systematically lower than the principal $\mu_{\theta\theta}$ that Yin’s phantom data seems to reveal.

Follow-up. Upgrade the constitutive core of `navier_solve_3d_tensor_mu` from the rank-2 μ_{ij} form to a full rank-4 $C_{ijkl}^{\text{eff}}(\mathbf{x})$ field. For incompressible neo-Hookean, C_{ijkl}^{eff} has a closed form in terms of the principal stretches $\lambda_i(\mathbf{x})$ and the pre-stress σ^{pre} (Ogden §6.1); no new material constants needed. Third-order Murnaghan coefficients (l, m, n) would be a further refinement on top, but those are not measured for the Sobh–Ehman phantoms and are expected to give smaller corrections than the rank-2→rank-4 upgrade.

Grid. $N = 48$ ($\Delta x = 3.75$ mm) is the current production size; PARDISO on 32 threads solves in ~ 200 s and uses ~ 25 GB (§7). Going to $N = 64$ –96 (matching Yin’s raw isotropic voxel) would multiply memory by ~ 2 – $8\times$ and would need either a bigger node reservation or an iterative solver with an appropriate preconditioner (block-Jacobi over voxels is the obvious first attempt).

12 Files and reproducibility

All under `/u/sobh/Eman_Richard/tsm.fno/`:

File	Purpose
<code>src/solver/helmholtz_fd_3d.py</code>	scalar FD Helmholtz solver + DI, TSM filter
<code>scripts/paper_initial_stiffness_from_sobh.py</code>	build G_{P1}, G_{P2} from paper constants
<code>scripts/paper_wave_sim_sobh_initial.py</code>	bottom-plate + DI forward run
<code>scripts/paper_wave_sim_sobh_mip.py</code>	multi-face + TSM MIP forward run
<code>scripts/paper_wave_sim_sobh_vector.py</code>	vector Navier + rotated tensor μ_{ij}
<code>results/paper_initial_stiffness_sobh/</code>	$G(\mathbf{x})$ npz + PNGs + summary
<code>results/paper_wave_sim_sobh_initial/</code>	bottom-drive results
<code>results/paper_wave_sim_sobh_mip/</code>	TSM MIP results, shell stats, histograms
<code>results/paper_wave_sim_sobh_vector/</code>	vector Navier results

Python environment: `/u/sobh/.conda/envs/mri_mrf_pytorch_env` (numpy, scipy, matplotlib).
Runs take a few minutes on a login node; no GPU required.

Version: this document was regenerated from live scripts on September 16, 2026. Any change to the scripts is captured in git under `github.com:nahilsobh/Eman_Richard`; the latest relevant commits are `0bb5fb5`, `ade3bfc`, `dc8d796`, `a0fbafd`.

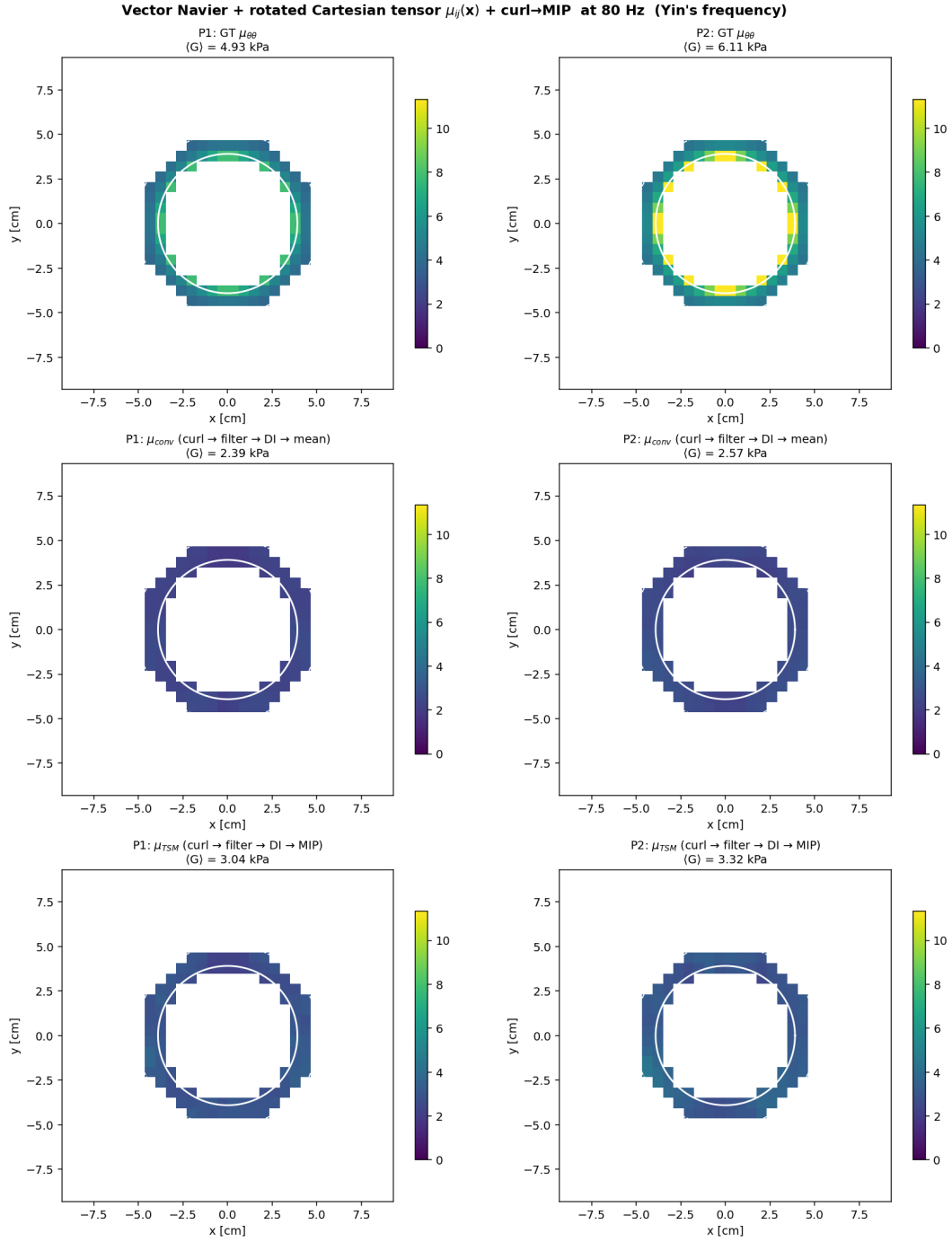


Figure 4: Shell heatmaps for the vector-Navier + tensor- μ_{ij} + curl→MIP pipeline at 80 Hz. Rows: GT $\mu_{\theta\theta}$ (top), μ_{conv} (middle), μ_{TSM} (bottom). Columns: P1 (left), P2 (right). All panels on the same colour scale. Grid is coarser (32^3 at 6 mm) than the scalar runs (60^3 at 3 mm) because vector Navier triples the DOF count.